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ADAM SHIRWA

PROFESSIONAL SUMMARY

Recent Honours Science graduate at the University of Waterloo with a Computing minor and certification in Genomic Data Science from Johns Hopkins University. Experienced in Python, R, Bash, and bioinformatics tools including BLAST, FastQC, and Prokka. Focused on applying computational methods to genomic and protein data analysis, with interests in bioinformatics, metagenomics, computational biology, and genomics-driven research roles.

EDUCATION & CERTIFICATIONS

BACHELOR OF SCIENCE **Sep 2021 – Dec 2025**
University of Waterloo ***Waterloo, Ontario***

Relevant Coursework: *Computational Biology, Genomics, Structural Bioinformatics, Genetics, Microbiology, Biostatistics and Experimental Design*

MINOR IN COMPUTING **Jan 2023 – Dec 2025**
University of Waterloo ***Waterloo, Ontario***

Relevant Coursework: *Applications in Software Development, Computer Applications in Business, Data Types, Structures and Management*

PYTHON IN GENOMIC DATA SCIENCE CERTIFICATION **March 2025**
Johns Hopkins University

SKILLS

Programming and Workflow: Python, Bash, R, SQL, Nextflow, Java, Racket, HTML,
Statistical Analyses: IBM SPSS Statistics, R
Bioinformatics and Genomics: BLAST, FastQC, Prokka, Artemis, Multiple sequence alignment, phylogenetics, sequence annotation
IDE's and Tools: Visual Studio Code, Jalview, Seaview, ACT, DrRacket, BBEdit, Microsoft 365 Applications

TECHNICAL EXPERIENCE

Evolutionary Conservation of BRCA1 Across Primate Species Reveals High Constraint in Functional Domains
Tools: COBALT, Jalview, NCBI, R, Excel, Python
Conducted comparative bioinformatics analysis of BRCA1 protein sequences across primate species to assess evolutionary conservation of functional domains.

- Curated BRCA1 protein sequences from NCBI and performed multiple sequence alignment using COBALT and Jalview
- Quantified conservation across RING, linker, and BRCT domains using pairwise identity matrices
- Constructed neighbour-joining phylogenetic trees with bootstrapping to evaluate evolutionary divergence
- Interpreted sequence conservation patterns in the context of functional constraint and disease relevance
- Validated alignment quality and consistency across tools to ensure robustness of downstream analysis
- Framed results in the context of functional constraint and relevance to disease-associated domains

LRU Cache with a capacity of 3 element pairs in Python via Visual Studio Code.

- Applied data structures and cache-management strategies to solve constrained resource problems.

Applied Bioinformatics & Data Skills

- Worked with biological sequence data, annotations, and alignments using public databases and command-line tools
- Applied computational methods to interpret biological meaning from genomic and protein datasets

Experimental Biology Background

- Hands-on experience with molecular biology and microbiology laboratory techniques through upper-year coursework
- Understanding of experimental design, sample preparation, and sources of technical variation relevant to genomic data analysis

HOBBIES & EXTRA-CURRICULAR ACTIVITIES

Formula 1, Gaming, Building PCs, Coding, Ice Hockey (Jan 2008 - Present)
VOLUNTEER EXPERIENCE: Big Brother Big Sister Mentor (Sep 2023-Oct 2024, Milton, Ontario)
ACHIEVEMENTS: Honour Roll (Jan 2021), University of Waterloo Merit Scholarship (Sep 2021), Term Distinction (December 2024, August 2025)